

Exp no 6 : Inference of Q Factor of Rectangular Cavity Resonator for TE₁₀₁ mode with the following inputs a) Dimension a in x axis b) Dimension b in y axis c) Dimension p in z axis

Test case 1:

```
clear;

clc;

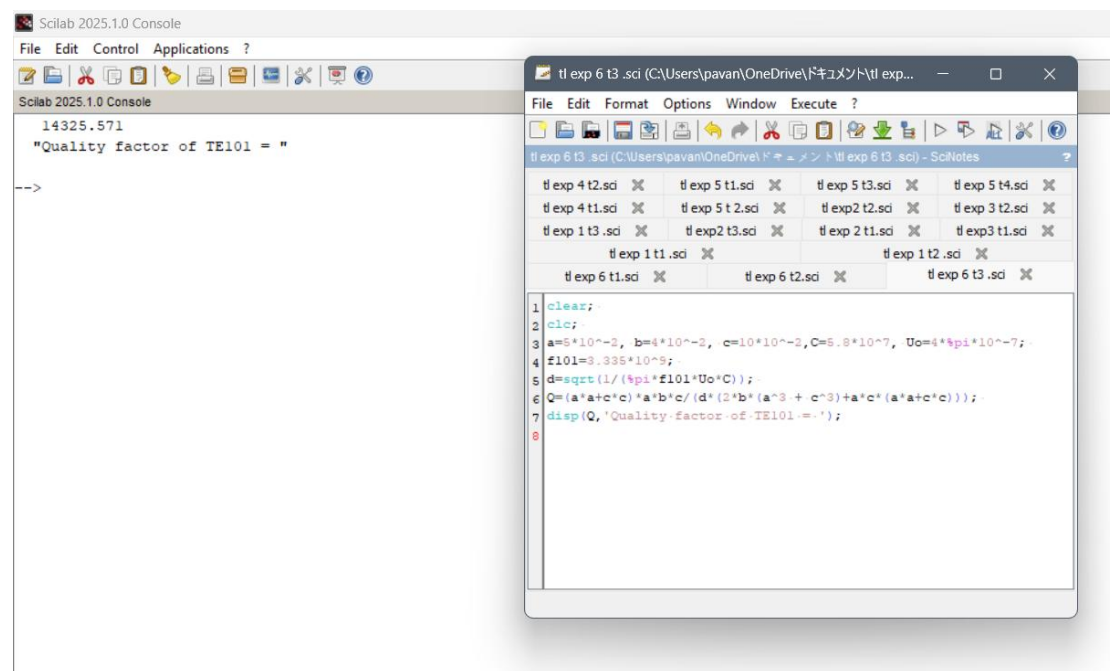
a=5*10^-2,b=4*10^-2,c=10*10^-2,C=5.8*10^7,Uo=4*%pi*10^-7;

f101=3.335*10^9;

d=sqrt(1/(%pi*f101*Uo*C));

Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));

disp('Quality factor of TE101 = ',Q);
```



Test case :

```
clear;
```

```
clc;
```

```
a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7;
```

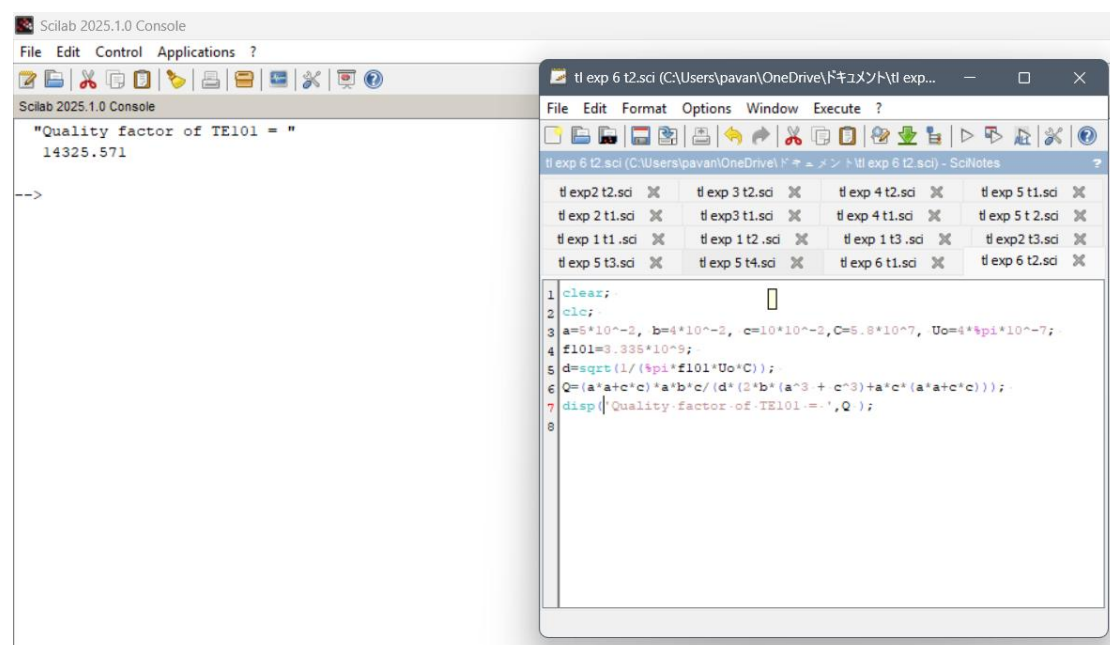
```
Uo=4*%pi*10^-7;
```

```
f101=3.335*10^9;
```

```
d=sqrt(1/(%pi*f101*Uo*C));
```

```
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
```

```
disp('Quality factor of TE101 = ',Q);
```



Test case 3:

```
clear;
```

```
clc;
```

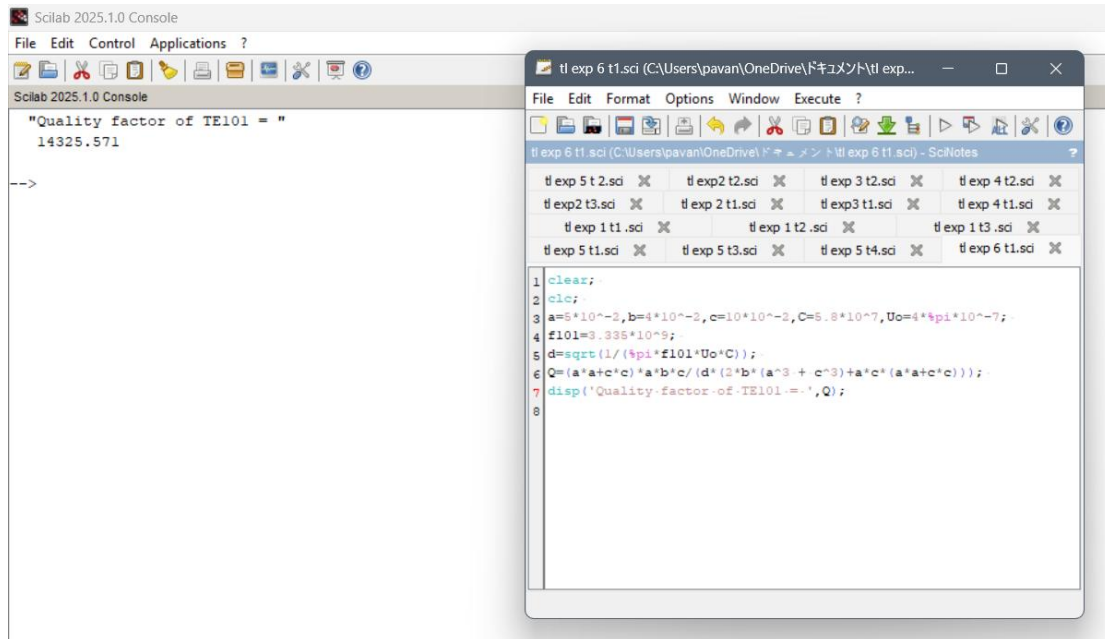
```
a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*%pi*10^-7;
```

```
f101=3.335*10^9;
```

```
d=sqrt(1/(%pi*f101*Uo*C));
```

```
Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));
```

```
disp(Q,'Quality factor of TE101 = ');
```



Test case 4 :

clear;

clc;

a=5*10^-2, b=4*10^-2, c=10*10^-2, C=5.8*10^7, Uo=4*%pi*10^-7;

f101=3.335*10^9;

d=sqrt(1/(%pi*f101*Uo*C));

Q=(a*a+c*c)*a*b*c/(d*(2*b*(a^3 + c^3)+a*c*(a*a+c*c)));

disp(Q,'Quality factor of TE101 =');

